Simulation exercise 4 - Electromagnetic plunger

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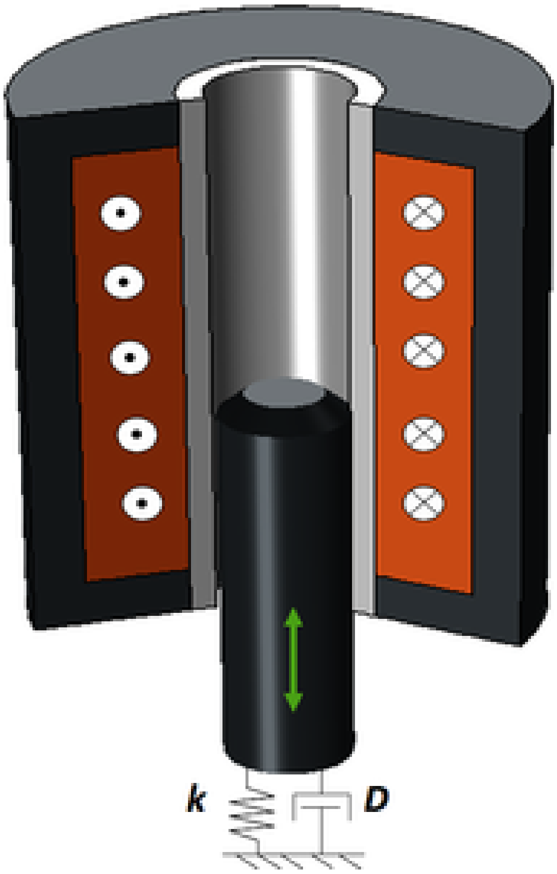

Simulation Exercise 4
FEM-modelling SE2860, 2025

Abstract

This project investigates the electromagnetic loading and dynamic response of a plunger-plate system using analytical methods and finite element simulations. In part (a), the electromagnetic force was evaluated analytically and numerically in COMSOL. Force integration over the plunger and coil gives nearly identical results, with a small systematic difference due to flux closing partly inside the coil. Compared to MATLAB's analytical model, numerical forces deviate by roughly 20%, mainly because the analytical approach assumes a point load, while the simulated load is distributed over the plunger's full contact area. This also explains why the numerical force-displacement response becomes stiffer at larger deformations. The analytical solution aligns well with small-displacement simulations but becomes inaccurate once deflections exceed ~0.08 mm. Parametric variations show that reducing the coil-plunger distance significantly increases electromagnetic force due to higher flux density in the air gap.

In part (b), the first natural frequency predicted by a mass–spring model (1396 Hz) was lower than the COMSOL result (1609 Hz), reflecting the increased stiffness introduced by the plunger's finite contact area. Time-dependent studies demonstrate that the excitation duration strongly affects the transient response: very short impulses excite mainly the lowest mode, while half-period impulses generate sustained oscillations when no damping is included. Introducing modal damping yields the expected decay in amplitude, with higher damping factors producing faster attenuation. The damping sensitivity also shows that the relative weighting of the first two modes influences the overall motion.

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1. Problem background and definition

Electromagnetic plunger systems are used in actuators, switches, and precision devices that convert electrical energy into controlled mechanical motion. Understanding their behaviour is essential for predicting performance and optimizing force output, response time, and structural durability. In this problem, a time-varying magnetic field interacts with an iron plunger to generate the force that moves it and induces stresses in an attached acrylic plate. Accurate modelling therefore requires capturing both the electromagnetic field and the resulting mechanical deformation, with emphasis on the magneto-mechanical coupling between them.

2. Model definition

2.1 Geometry and materials

Figure 1 shows the model geometry, and Table 1 lists the parameters. The iron plunger is attached to a clamped acrylic support plate, and both materials are assumed isotropic and linearly elastic. The solenoid's copper wire is insulated with a material treated as air. When parameter ranges were given in Table 1, the upper value was used.

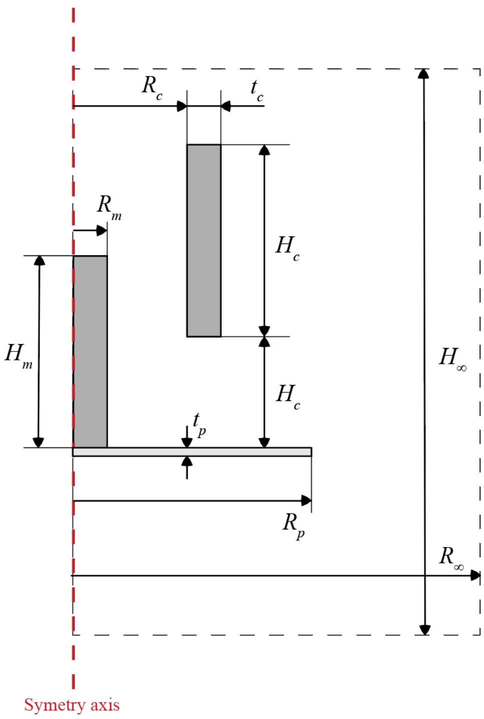


Figure 1: *Axisymmetric cross section of electromagnetic plunger, with variables of dimensions and geometry.*

Table 1: *Parameters used for the assignment.*

Parameter	Value	Dimension/Unit
R_p	5	mm
t_p	0.2	mm
R_c	2	mm
t_c	0.7	mm
H_c	5	mm
H_0	2.5	mm
R_m	0.7	mm
H_m	5.5	mm
N_{turns}	4000	-
A_{cross}	$10*10^{-6}$	m^2
σ	$6*10^7$	S/m
ρ_{acryl}	1190	kg/m^3
E_{acryl}	3.2	GPa
v_{acryl}	0.35	-
ρ_{iron}	7500	kg/m^3
E_{iron}	160	GPa
v_{iron}	0.24	-
B_{flux}	1.4	T

2.2: Relevant equations

2.2.1: Electromagnetism Equation

The following are the governing equations used in electromagnetism in COMSOL and the analytical solutions. The principal equation used in FEM analysis are Maxwell's equations, as seen in Eq. 1-4:

Gauss's law: $\nabla \cdot \boldsymbol{D} = \rho$ *(1)*

Faraday's law: $\nabla \times \boldsymbol{E} = -\partial \boldsymbol{B} / \partial t$ *(2)*

Gauss's magnetic law: $\nabla \cdot \boldsymbol{B} = 0$ *(3)*

$$Maxwell\text{–}Ampère's\ law: \qquad \nabla \times \boldsymbol{H} = \boldsymbol{J} + \partial \boldsymbol{D}/\partial t \qquad (4)$$

Where ∇* is the divergence operator, $\boldsymbol{D}$ is the electric flux density, $\boldsymbol{\rho}$ is the free electric charge density, $\nabla\times$ is the curl operator, $\boldsymbol{E}$ is the electric field intensity, and $\boldsymbol{B}$ is the magnetic flux density, $\boldsymbol{H}$ is magnetic field intensity, $\boldsymbol{J}$ is the electric current density, $\boldsymbol{D}$ is the same as in Eq 1. The above equations together form a partial differential equation that can be solved in COMSOL. To make the simulations more stable a magnetic vector potential, $\boldsymbol{A}$, is introduced that automatically satisfies the relationship of Gauss magnetic law (Eq. 3). See Eq. 6.

$$\nabla \cdot \boldsymbol{B} = 0 \qquad (5)$$

$$\boldsymbol{B} = \nabla \times \boldsymbol{A} \qquad (6)$$

The relationship between the conductivity and the electrical current density is governed by Ohm's law:

$$\boldsymbol{J} = \sigma \boldsymbol{E} \qquad (7)$$

Where σ is the electrical conductivity. A constitutive relationship can also be established between $\boldsymbol{B}$ and $\boldsymbol{H}$, as seen in Eq. 8:

$$\boldsymbol{B} = \mu_0 \mu_r \boldsymbol{H} \qquad (8)$$

Where $\mu = \mu_0 \mu_r$ is the absolute permeability of the material.

2.2.2: Solid mechanics equations

The following section will feature an explanation to how electromagnetic forces are calculated in COMSOL. The volume force applied from electromagnetism is defined by the Lortenz force density, Eq. 9.

$$\boldsymbol{f_m} = \boldsymbol{J} \times \boldsymbol{B} \qquad (9)$$

Where f_m is the Lorentz force density. The total force acting on the body can be calculated using a volume integral. The surface traction felt by a non-conducting body or when air is present et cetera is calculated

$$\boldsymbol{\sigma_m} = \boldsymbol{B} \otimes \boldsymbol{H} - \frac{1}{2}(\boldsymbol{H} \cdot \boldsymbol{B})*I \qquad (10)$$

In COMSOL, the stress tensor (Eq. 10) can then be transformed into the traction vector and then integrated over a desired closed surface. The traction vector is calculated as:

$$\boldsymbol{t_m} = \boldsymbol{m} \cdot \boldsymbol{n} \qquad (11)$$

This is later implemented into COMSOL and may be used to calculate, among other things, the displacement.

To calculate the magnetic force acting on a body, $\boldsymbol{t}_m$ is integrated over the surface of the body according to Eq. 12.

$$\boldsymbol{F} = \int_S \boldsymbol{\sigma}_m \cdot \boldsymbol{n} \, dS \qquad (12)$$

Where S is the entire surface of the object we want to calculate the force on.

2.3: Boundary conditions.

For both cases, a symmetry boundary condition was applied to the left boundary, along with a fixed boundary condition for the acrylic plate. The magnetic isolation boundary condition was set for the far boundaries, as shown in Figure 2.

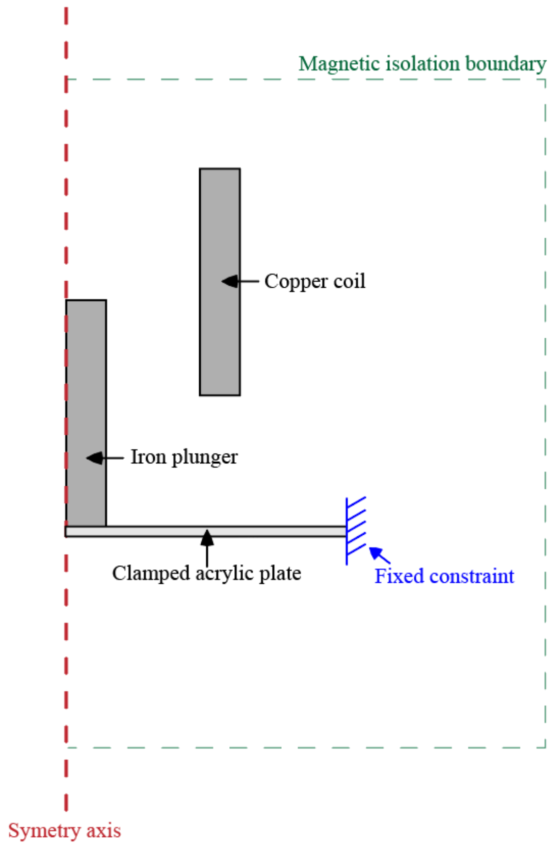


Figure 2: *Boundary conditions used for both cases.*

3. Numerical model

3.1: Mesh used for case (a) and (b)

The numerical model used parameter values from Table 1 and geometry from Figure 1. A free triangular mesh was applied to the plunger, with a maximum element size of 0.0005/*N_mesh*. A distribution was applied around the plunger and the acrylic plate (blue lines in Figure 3), set to 10*N_mesh.

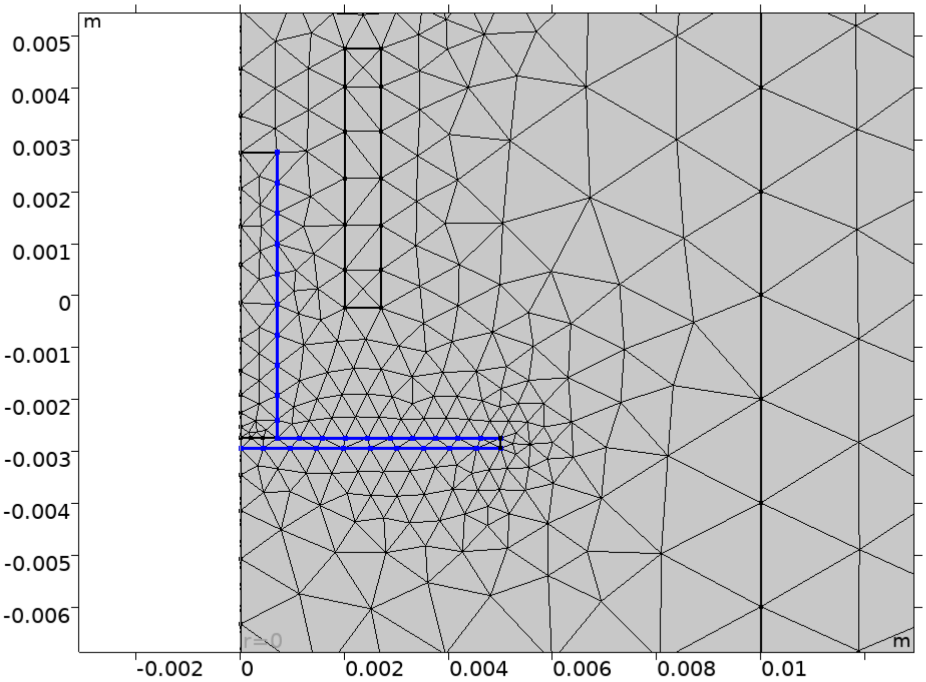
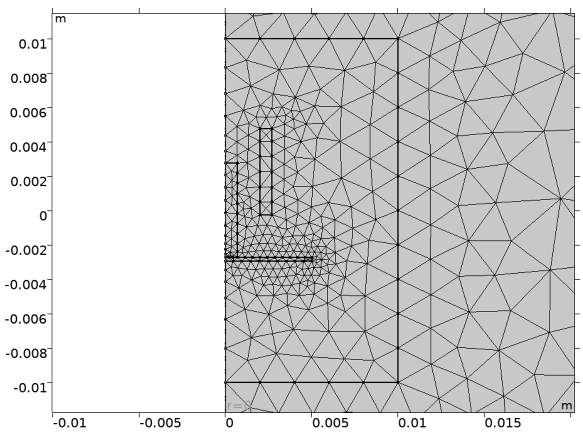


Figure 3: *Distribution of mesh around border of plate and plunger in blue.*

A rectangle was introduced for smaller meshing around the area of interest, with a maximum element size of 0.002/*N_mesh* inside, right borders set to distribution with 10**N_mesh* and upper and lower boundary set to distribution with 5**N_mesh*, as shown in Figure 4. The remaining mesh had a maximum element size of 0.01/*N_mesh*.



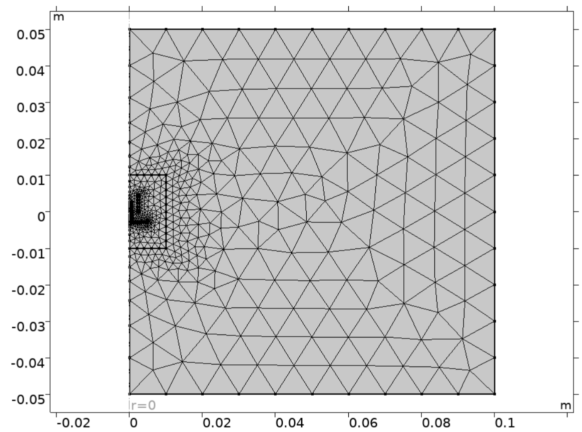


Figure 4: *Left: Mesh of area around coil, plate and plunger. Right: Full mesh of the entire model during simulation. N_mesh = 1.*

3.2: Mesh convergence

A mesh convergence plot was created by varying *N_mesh* from 1 to 40 in increments of 3 plotting against the average displacement of the acrylic plate. As seen in Figure 5, a *N_mesh* of around 16 was deemed sufficient and used in case (a) and (b) as further increases led to increased computational cost.

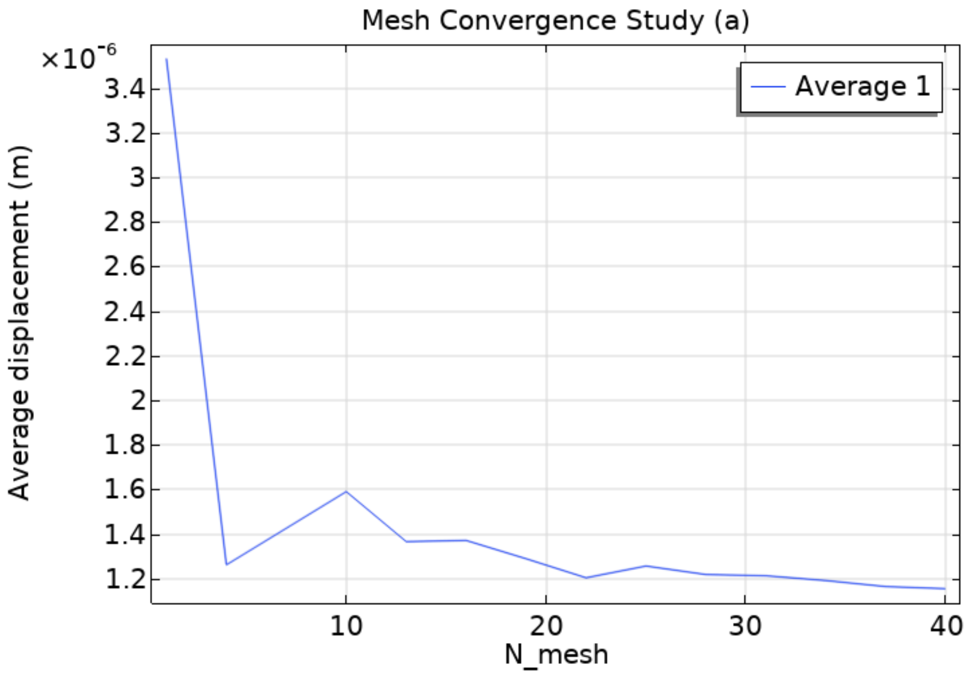


Figure 5: *Mesh convergence study for case (a)*

4. Results and Discussion

4.1: Result for case (a)

4.1.1: Analytical solution

For analytically estimating the relation between force on the plate and deflection of it, the elementary solutions in (Alfredsson, 2014, Table 31.5) were used. The first and third case in the table were combined to facilitate fixed boundaries.

The angle at the edges is computed by Eq. 13 should be equal to zero with fixed constraints.

$$\theta(r = R_p) = \frac{MR_p}{D(1+\nu_{acryl})} + \frac{FR_p}{4\pi D(1+\nu_{acryl})} = 0 \quad (13)$$

$$\text{where } D = \frac{E_{acryl} t_{acryl}^3}{12(1-\nu_{acryl}^2)} \quad (14)$$

Where M is the reaction torque from the fixed constraints, F is the force from the plunger on the plate. Eq. 13 solved gives $M = -\frac{F}{4\pi}$ which can be plugged into case 1 and combined with case 3 to evaluate the displacement.

$$\delta = \frac{-\frac{F}{4\pi} R_p^2}{2D(1+\nu_{acryl})} + \frac{FR_p^2}{8\pi D} \frac{3+\nu_{acryl}}{2(1+\nu_{acryl})} = \frac{FR_p^2}{16\pi D} \quad (15)$$

To calculate the force F analytically, Eq.12 is used with $\boldsymbol{\sigma_m}$ according to Eq.10. Since the force only will be in z-direction from the top of the plunger, the formula can be rewritten as

$$\sigma_{m,z} = B_z H_z - \frac{1}{2} B_z H_z = \frac{1}{2} B_z H_z, \qquad (16)$$

With $B_z = B_{flux}$ and $H_z = \frac{N_{turns} I}{H_c}$ (Electronics-tutorials, 2025) with I being the current. The force acting on the plunger is thus,

$$F = \int \boldsymbol{\sigma_m} \cdot \mathbf{n_z}\, dS \qquad (17)$$

Since the sides of the plunger has a normal orthogonal to $\sigma_{m,z}$, Eq. 17 can be reduced to

$$F = \int_0^{2\pi} \int_0^{R_m} \frac{B_{flux} N_{turns} I}{2H_c} r\, dr d\theta = \frac{\pi R_m^2 B_{flux} N_{turns} I}{2H_c} \qquad (18)$$

This force is calculated using MATLAB and plotted against current together with the numerical evaluations of the force (Fig. 6 and 7).

4.1.2: Force calculation method analysis

Figure 6 shows a force-current plot for the two manners of integrating the force, and also the analytical solution of the same value.

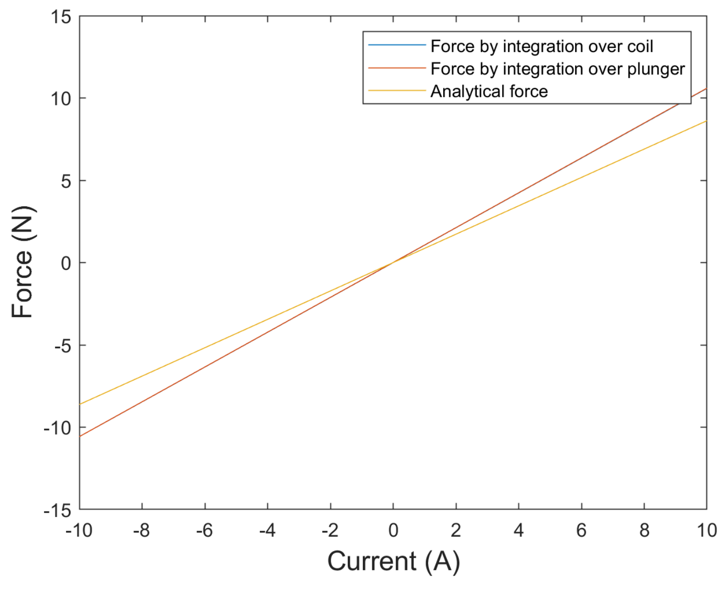


Figure 6: *Force-Current graph for the three manners of calculating the force of the plunger. For -10 to +10 Ampere.*

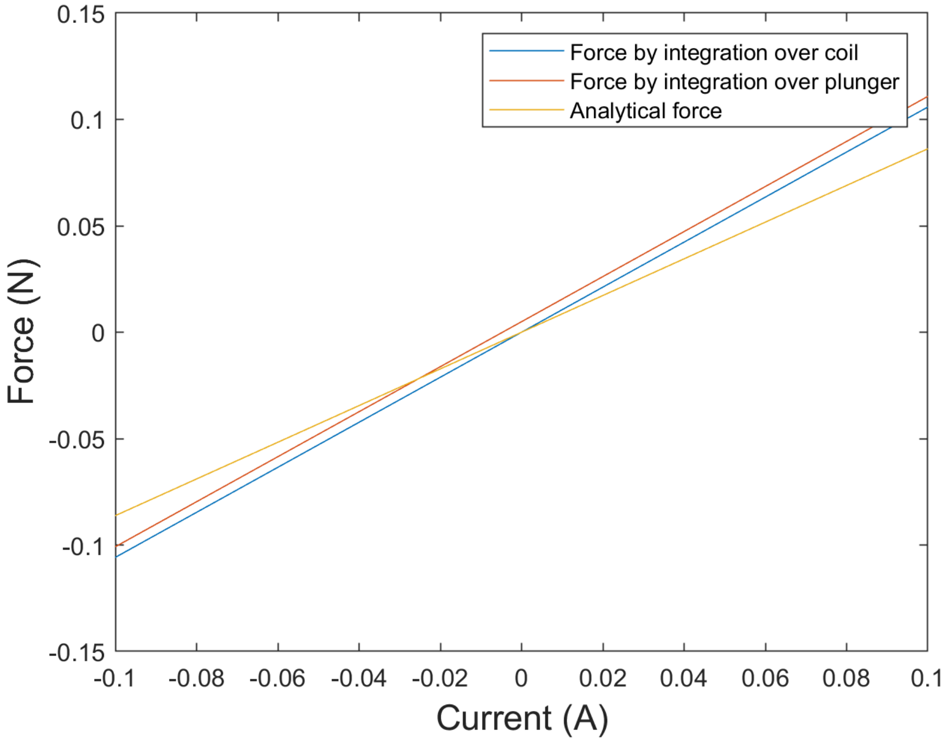


Figure 7: *Zoomed in version of Figure 6.*

Figure 7 shows a zoomed-in version of Figure 6. The force integrated over the coil and integration over the plunger are similar enough that the force integration over the coil is difficult to distinguish in Figure 6. The coil-integrated force is slightly smaller than the plunger-integrated force because part of the magnetic field produced by the coil closes its flux path inside the coil itself, rather than across the air gap. As a result, not all of the coil's magnetic field contributes to pulling on the plunger, while the plunger "samples" a slightly stronger and more focused portion of the flux in the gap. This leads to a small but consistent difference between the two integrals. This difference is minor and can be ignored, especially at higher currents. A comparison with MATLAB's analytical solution shows a deviation of about 20% from the numerical results. Aside from the 0-current case, where both methods correctly predict zero force. The absolute difference increases with current, but the relative error remains roughly constant. Finally, integration over the coil yields a negative force, but since the plotted force is the one acting on the plunger, the sign is reversed to positive.

4.1.3: Analysis and results of analytical and numerical

The comparison between the analytical and numerical solutions was performed in MATLAB. In Figures 8 and 9, where the force is computed by integrating over the plunger, the analytical model becomes less accurate at larger displacements. This is because the analytical model assumes small deformations.

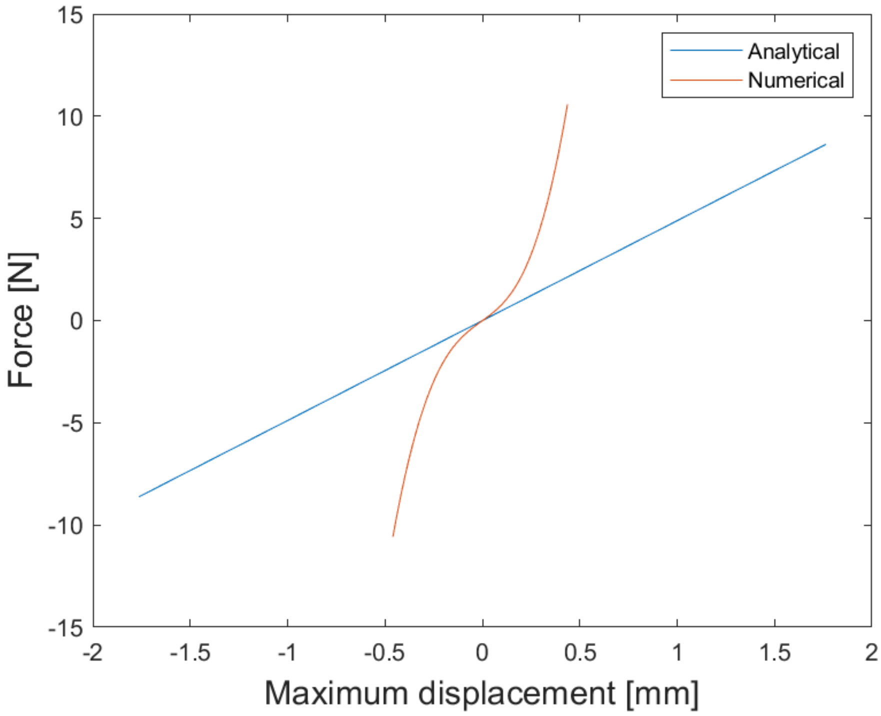


Figure 8: *Force-displacement plot for numerical and analytical.*

In Fig. 9, it is seen that for small deformations shows a higher stiffness. This deviation arises because the analytical solution assumes a point load, while the numerical model where the plate is connected to the plunger over a non-negotiable area, effectively making the numerical response stiffer than the analytical solution.

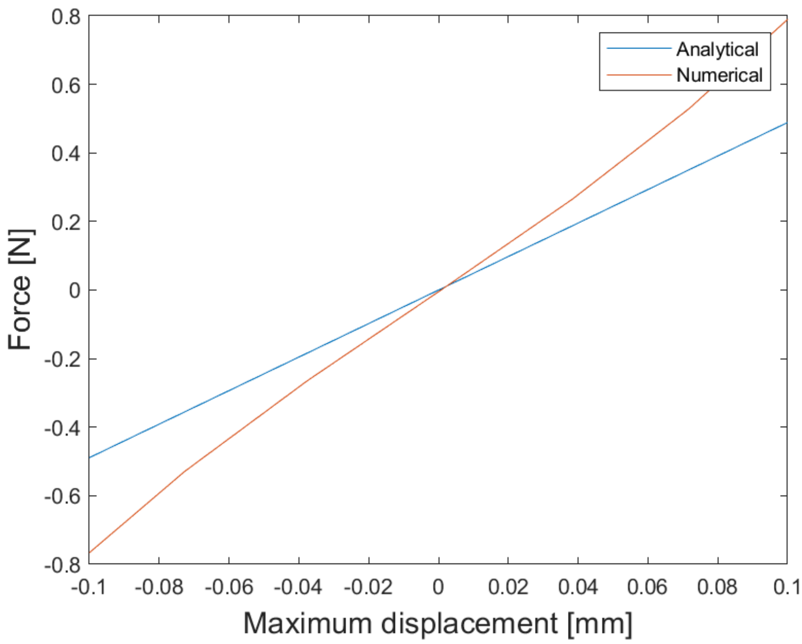


Figure 9: *Zoomed in force-displacement plot for analytical and numerical solution.*

4.1.4: Comparison of large and small deformation

Figure 10 is similar to Figure 8, with the small-deformation curve closely matching the analytical solution in both form and value. This shows that the analytical model aligns better with the linear small-deformation response than with the large-deformation case. Figure 11 further indicates a relative difference of about 10% when the large-deformation displacement reaches roughly 0.08 mm, after this point the analytical solution is largely inaccurate.

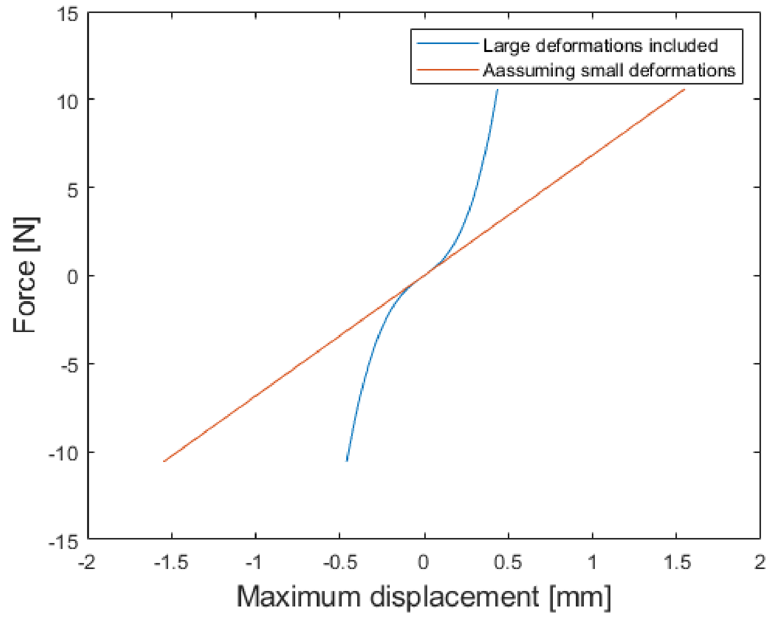


Figure 10: *Force-displacement graph for assuming large and small deformation.*

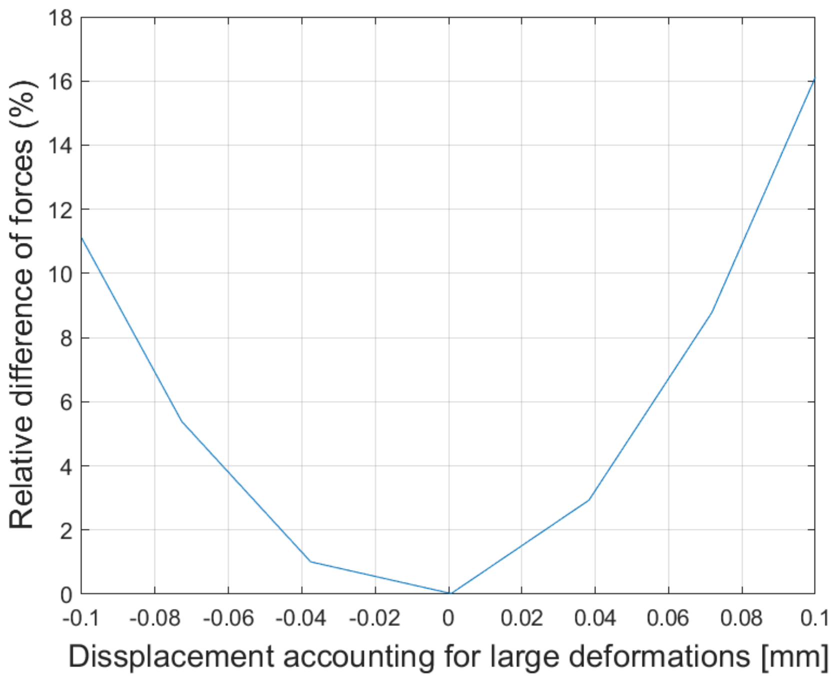


Figure 11: *Relative difference between large-small deformation plot.*

4.1.5: Displacement vs electric current

Figure 12 shows the same relationship as in Figure 10, i.e. that the analytical solution is similar when the current and thus the displacement is small and gets more inaccurate when the current and thus the displacement is increased. Increasing the current will increase the displacement.

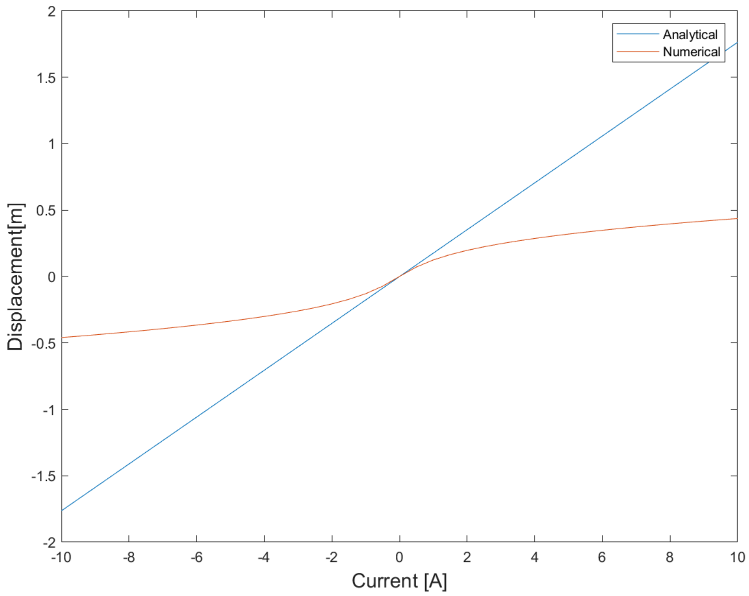


Figure 12: *Displacement-current graph for numerical and analytical solution.*

4.1.6: Varying distance between coil and plunger

By using parametric sweep in COMSOL the position of the coil is adjusted by increasing the radius *Rc,* from the symmetry axis. A step length of 0.1 mm is used from 1mm to 2mm. The electromagnetic force can then be derived by using the *mf.Forcez_plunger* and running the simulation with the different geometries, as seen in Figure 13.

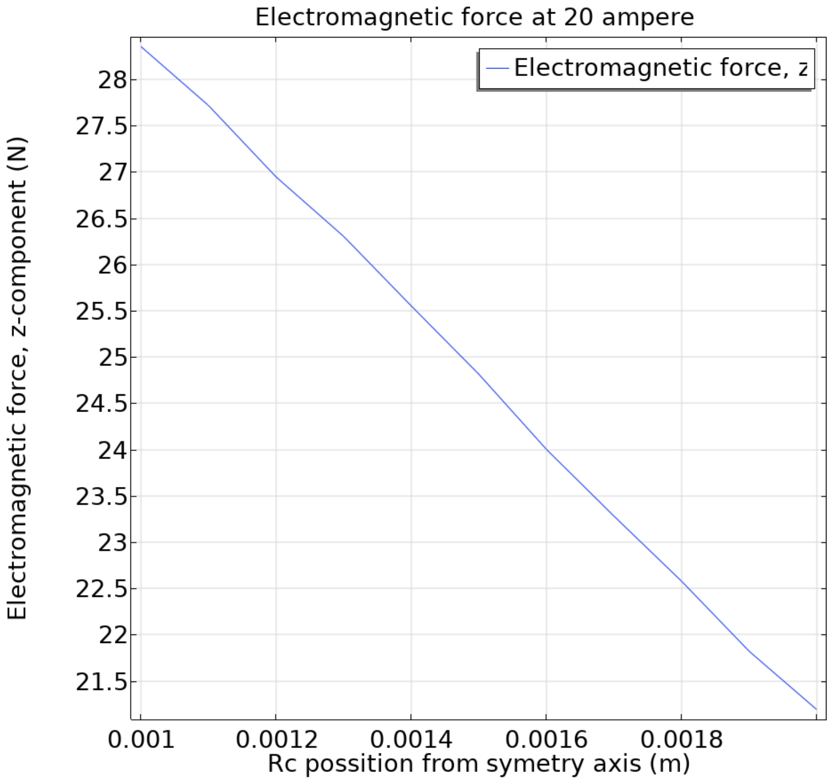


Figure 13: *Electromagnetic force with varying distance of coil and plunger. Electric current is 20 A.*

As expected when the coil approaches the magnet the force increases due to the magnetic field repelling the magnetic core. Figure 14 shows the concentration of flux density norm for the different positions of the copper coil.

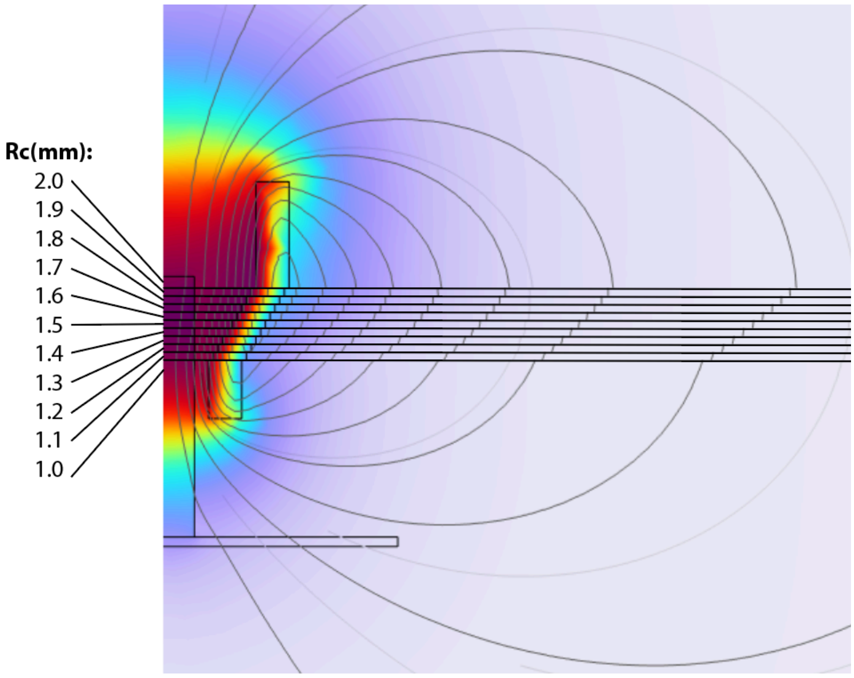


Figure 14: *Illustration of electromagnetic flux density norm with varying position of coil.*

4.2: Result for case (b)

The first natural frequency was calculated using a mass-spring model according to Equation 19 that was given in the help instructions.

$$f_{1,analytic} = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \qquad (19)$$

Where $k = \frac{16\pi D}{R_p^2}$ is the stiffness of the spring according to Eq. 19 and $m = \rho_{iron}\pi R_m^2 H_m$ is the mass of the magnet. This gives $f_{1,analytic}=1396.4\ Hz$.

The five lowest natural frequencies were calculated using COMSOL and yielded the results in Table 2.

Table 2: *Mode and natural frequency.*

Mode [-]	Frequency [Hz]
f_1	1609.5
f_2	18591
f_3	49906
f_4	95520
f_5	153380

The analytical lowest frequency (1396.4 Hz) is lower than the numerical result (1609.5 Hz). This is expected: the analytical model assumes a point force, whereas the numerical model accounts for the magnet being fixed to the plate over a large contact area. This larger fixed surface increases the effective stiffness, raising the natural frequency.

The current in the coil was set to the function shown in Figure 15 using “piecewise function” in COMSOL.

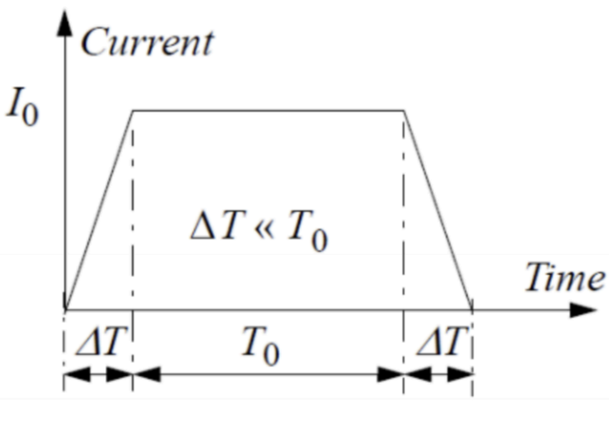


Figure 15: *Current as a function of time*

A time dependent study was conducted to model the movement of the plate in time from the current impulse. This was done using the same mesh and COMSOL settings as in (a), except excluding geometric non-linearities. Therefore the current I_0 was set to 0.5 A for all simulations in (b) since it was seen in (a) that geometric non-linearities had a strong effect when the current had higher amplitude. The time ΔT was set to $0.01\frac{1}{f_1}$ and the time T_0 was varied as seen in Figure 16 to $T_0 = c\frac{1}{f_1}$

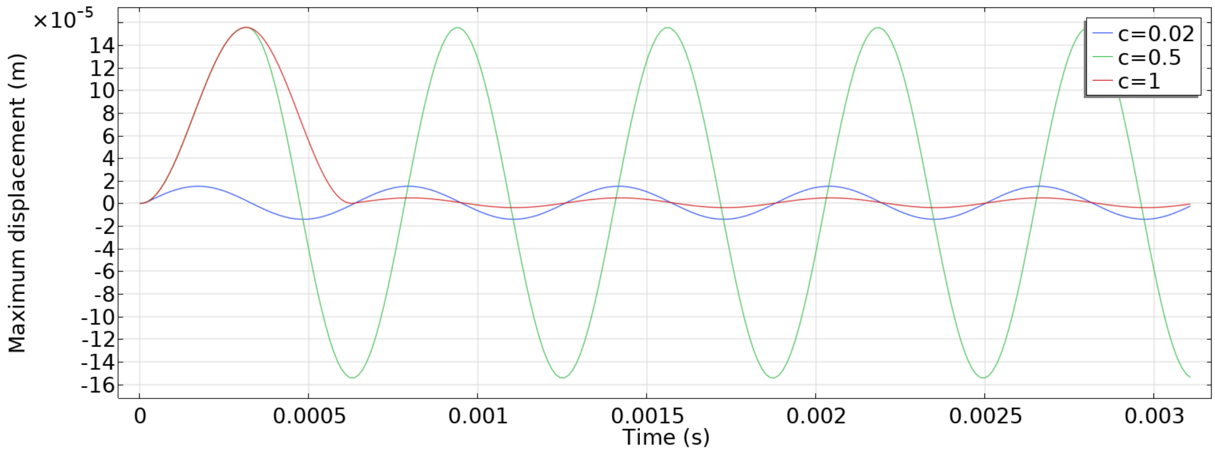


Figure 16: *displacement as function of time for different T_0*

T_0 was varied from 0 to 1, but values c < 0.02 caused unresolved convergence issues. Figure 16 shows that a short impulse (c=0.02) excites motion dominated by the lowest natural frequency. For c=0.5, the current turns off after half a period, so the release aligns with the plate's natural frequency, producing sustained oscillations since no damping is included. For c=1, T_0 equals one full period, meaning the release is out of phase and the displacement should ideally return to zero. The remaining oscillations in the red curve likely stem from the finite time step ΔT.

Damping was inserted under solid mechanics, and different damping (d) were investigated for the first two natural frequencies. The damping factor d_1 for the natural frequency f_1 and damping factor d_2 for f_2. These follow the criteria in Eq. 18.

$$\frac{f_1}{f_2} < \frac{d_1}{d_2} < \frac{f_2}{f_1} \qquad (18)$$

With numbers:

$$0.0866 < \frac{d_1}{d_2} < 11.551 \qquad (19)$$

Different combinations of the damping factors 0.01, 0.5 and 1 are shown in Table 3 with combinations that do not fulfill Eq. 19 marked red.

Table 3: *Combinations of d_1 and d_2 with d_1/d_2.*

	d_2=0.01	d_2=0.5	d_2=1
d_1=0.01	1	0.2	0.1
d_1=0.5	50	1	0.5
d_1=1	100	2	1

The resulting displacement-time graphs for the allowed combinations are shown in Figure 16 for *c=0.5*, this influences the plate so that it oscillates with lower and lower amplitude over time which is to be expected. This effect is higher for higher damping factors as seen for $d_1=d_2$. It is also seen that it matters what frequency the damping factors are related to, otherwise ($d_1=1$; $d_2=0.5$) would yield the same results as ($d_1=0.5$; $d_2=0.1$), which they do not according to Fig. 16.

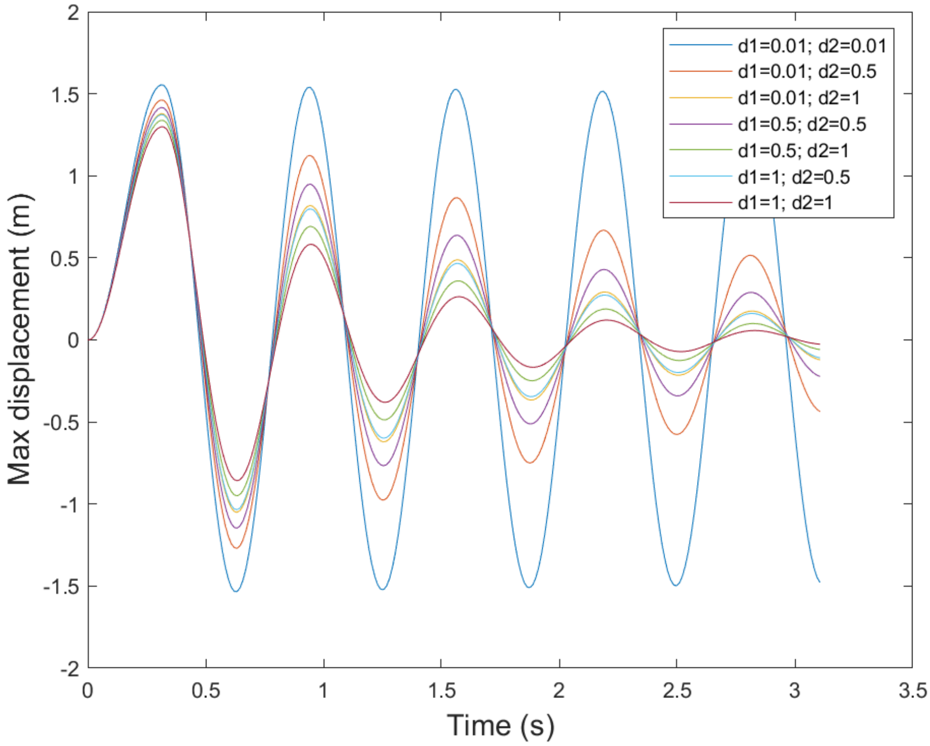


Figure 16: *Max displacement as function of time for different damping combinations.*

5. Conclusions

5.1: Case (a)

- The analytical plate-deflection model matches the numerical results only for small displacements; at larger deformations it becomes increasingly inaccurate.
- Numerical stiffness is higher because the load is distributed over the plunger, unlike the analytical point-load assumption.
- The analytical magnetic-force estimate differs by about 20% from numerical results, with a nearly constant relative error over the current range.
- Force integration over coil and plunger gives similar trends; the coil method is slightly lower due to internal flux closure.
- Higher current increases force and displacement, but the analytical model is only reliable at low current levels.
- Reducing the coil–plunger distance increases the electromagnetic force, confirmed by higher flux density in the air gap.

5.2: Case (b)

- The analytical first natural frequency (1396.4 Hz) is lower than the COMSOL result (1609.5 Hz) because the analytical model assumes a point attachment, while the numerical model includes a larger fixed contact area, increasing stiffness.
- A time-dependent study with a low current amplitude (0.5 A) was used to avoid geometric nonlinearity effects observed at higher currents.
- Short current impulses excite mainly the first natural frequency, while impulses aligned with half a period (c = 0.5) produce sustained oscillations due to lack of damping.
- When the impulse duration corresponds to a full period (c = 1), the system ideally returns to zero displacement.
- Valid damping-factor combinations must satisfy the ratio condition in Eq. 18; simulations confirm that higher damping reduces oscillation amplitude more quickly.

6. AI

ChatGPT was used to consult the meaning of certain sentences in the laboratory instructions, and summarize longer paragraphs. It was also used to correctly cite sources used in the reference section.

7. References

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Electronics Tutorials. (2025, November 29). *Electromagnets*. Electronics-Tutorials. https://www.electronics-tutorials.ws/electromagnetism/electromagnets.html

Faleskog, J. (2025, November). FEM modelling [Seminar]. KTH Royal Institute of Technology, Stockholm, Sweden.

8. Appendix

Appendix A: Eigenmodes for all natural frequencies

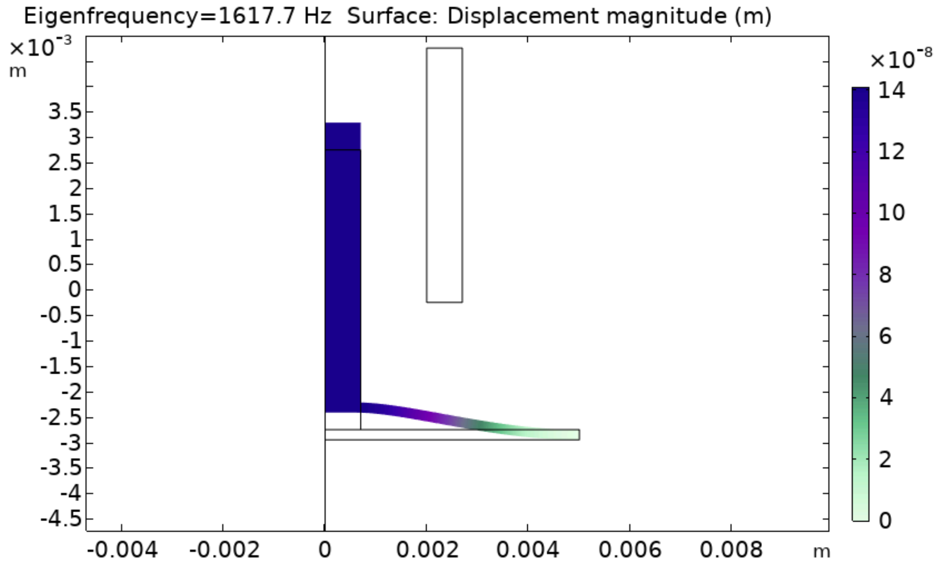

Figure A: *Eigenfrequency 1617 Hz, surface displacement magnitude (m)*

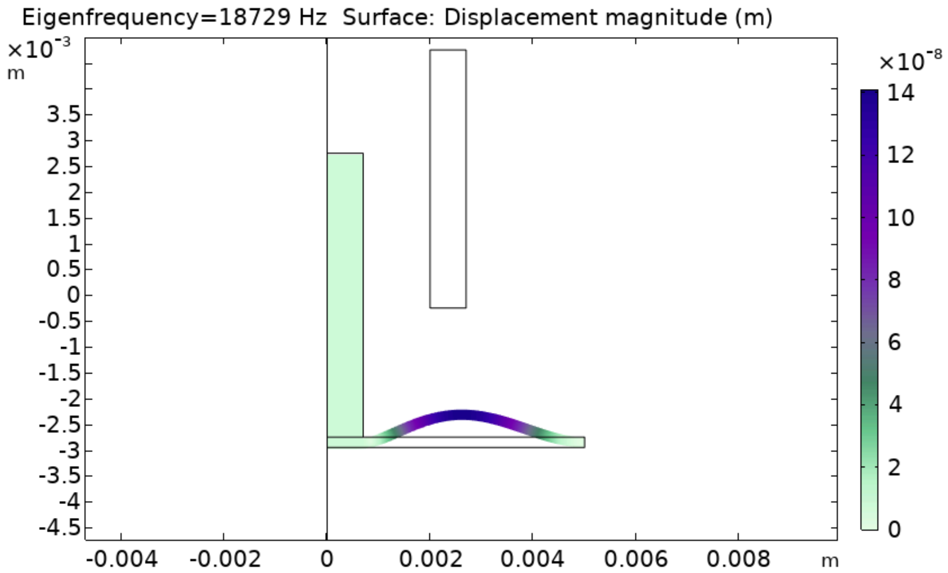

Figure B: *Eigenfrequency 18729, surface displacement magnitude (m)*

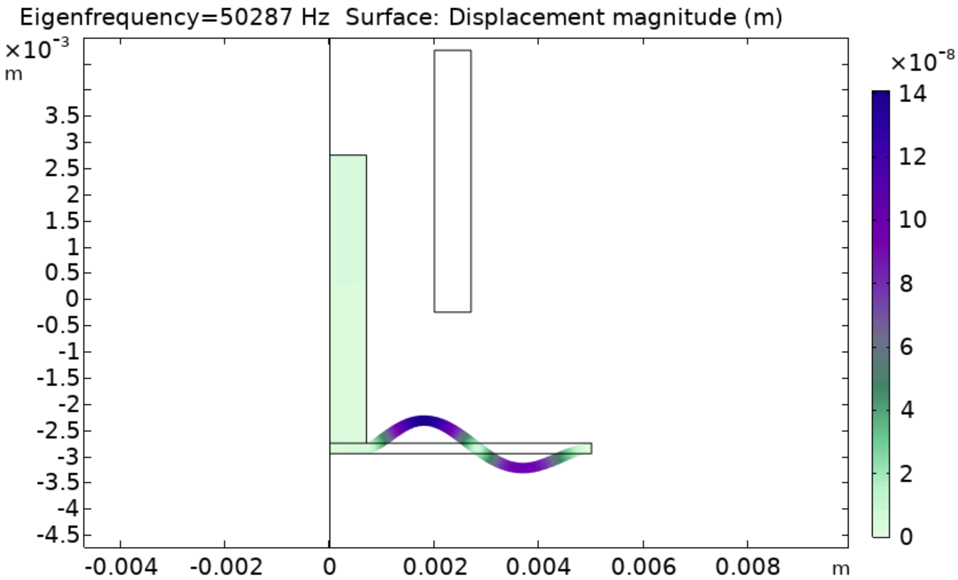

Figure C: *Eigenfrequency 50287, surface displacement magnitude (m)*

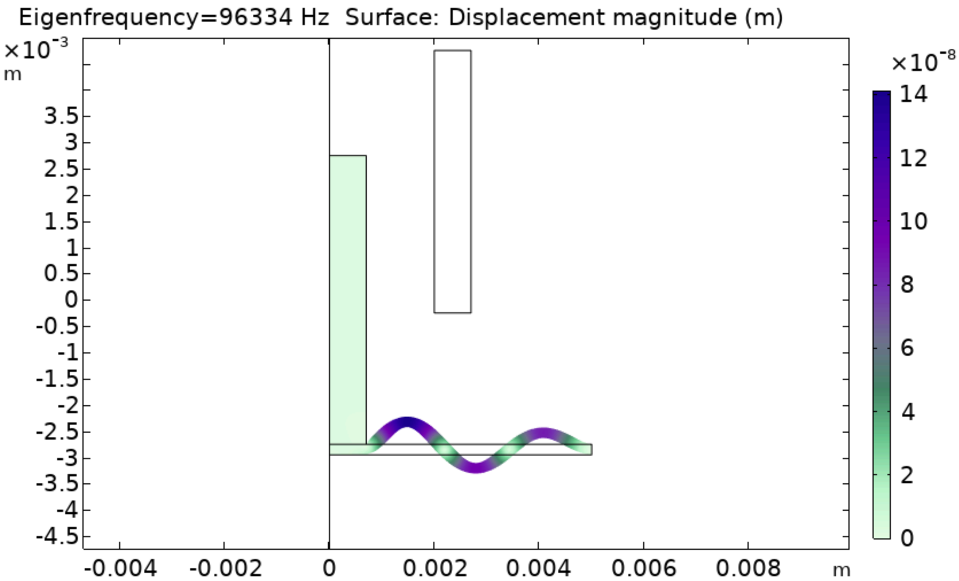

Figure D: *Eigenfrequency 96334 Hz, surface displacement magnitude (m)*

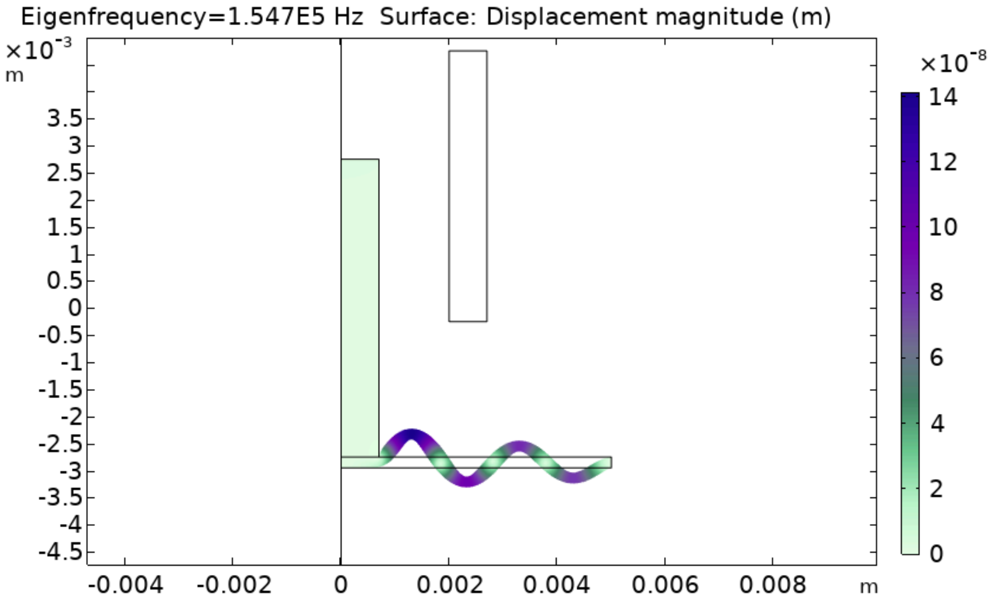

Figure E: *Eigenfrequency is 1.547E5 Hz, surface displacement magnitude (m)*

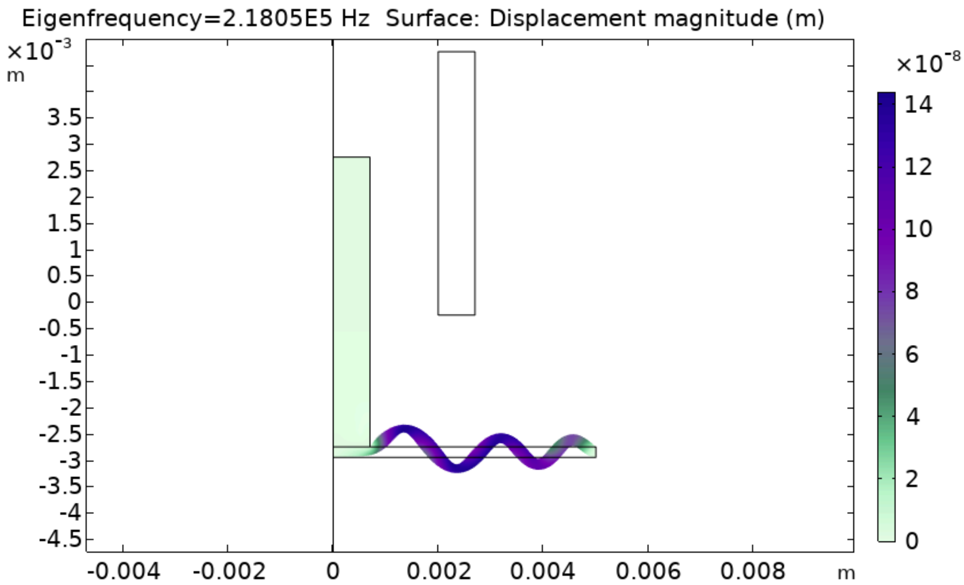

Figure F: *Eigenfrequency is 2.1805E5 Hz, surface displacement magnitude (m)*